

# TAHSIN TOUHEED IVAN

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## EDUCATION

### University of Dhaka

*B.Sc. in Electrical and Electronic Engineering (4th year)*

CGPA: 3.73 / 4.00

Dhaka, Bangladesh

Sept 2022 – Dec 2026 (expected)

### Notre Dame College

*Higher Secondary Certificate (Science)*

GPA: 5.00 / 5.00

Dhaka, Bangladesh

Aug 2019 – July 2021

### Ideal School & College

*Secondary School Certificate (Science)*

GPA: 5.00 / 5.00

Dhaka, Bangladesh

Jan 2009 – July 2019

## RESEARCH

### Materials Science Research Laboratory (MSRL DU)

Jan 2026 – Present

*Undergraduate Researcher | Supervisor: Prof. Dr. Imtiaz Ahmed*

Interests: ML-guided electrode design and materials discovery for energy storage, with broader curiosity in nanomaterials characterization and micro/nano-engineered devices.

### Current Work — Machine-Learning & DFT-Guided Electrode Design for Oxide-based Supercapacitors: From Data Mining to Experimental Validation

- Building a computation-guided closed-loop pipeline for perovskite-oxide ( $ABO_3$ ) based supercapacitor electrodes. Stage 1 was targeted towards curating a literature-mined descriptor dataset of 257 compositions (across 57 published experimental works) spanning chemistry, structure, processing and testing-condition features, and built a classifier model ensemble to rank compositions by propensity for top-tier specific capacitance; SHAP/PDP explainability identified the most dominant drivers, yielding a ranked shortlist of undoped and doped candidates.
- A stability screening model narrows the shortlist before first-principles validation, concentrating compute on the most viable candidates; surviving compositions enter a DFT workflow establishing the true ground state and confirming mechanical (elastic constants, Born criteria) and dynamical (phonon) stability, followed by charge-storage descriptors — quantum capacitance from the density of states, ion adsorption energy with Bader charge transfer, and volume expansion on ion insertion.
- Stage 2 targets experimental validation: screened candidates synthesized toward sub-200 nm particles for near-surface Faradaic utilisation, electrodes fabricated by spray pyrolysis or laser-induced graphene, structurally characterized (XRD, BET, Raman, XPS) and electrochemically tested (CV, GCD, EIS), with measured Csp fed back to refine the model; closing the loop.

## LEADERSHIP

### IEEE SIGHT Student Branch, DU

Feb 2025 – Present

#### Chair

April 2026 – Present

- Led the team that secured the IEEE HTB Tech4Good 2026 grant (USD 4,739) for *SmartDrip*, an IoT-enabled drip irrigation system for urban rooftop plantation; currently leading project execution end-to-end.

#### Program Coordinator

Feb 2025 – March 2026

- Coordinated and organized the *Machine Learning Fundamentals* workshop for undergraduate students — managed logistics, content planning, and participant engagement.
- Led a hands-on session in the *IEEE Day Outreach Programme on Microcontrollers and Their Utilization* — jointly organized by IEEE SB University of Dhaka, IEEE EDS SBC DU, and IEEE WIE AG SB DU at Nawab Faizunnesa Govt Girls High School, Cumilla — for 50+ high school students.

### Optics and Photonics Club

Jan 2024 – April 2025

## Vice President

Jan 2024 – April 2025

- Delivered talks and project showcases (LFR, Holography, etc.) at the outreach program *Envisioning Harmony Between Human and Nature*, St. Joseph Higher Secondary School, Dhaka — in collaboration with the Optica Bangladesh Student Chapter, University of Dhaka, and Scintilla Science Club.

## PROJECTS

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**SmartDrip:** IoT-Enabled Drip Irrigation System for Urban Rooftop Plantation | *Embedded Systems* 2026 – Ongoing

**Awarded:** *IEEE HTB Tech4Good 2026 Grant (USD 4,739)*

- Leading an IEEE SIGHT SB DU team developing an IoT-enabled drip irrigation system aimed at making urban rooftop plantation water-efficient and low-maintenance for Old Dhaka residential area — addressing freshwater stress and food-resilience challenges in dense urban areas.
- Proposal selected for funding under the IEEE Humanitarian Technologies Board’s Tech4Good 2026 call; currently directing system design, sensor-actuator integration, and field-deployment planning.

**LC-Based Capacitive MEMS Pressure Microsensor** | *Python, scikit-learn* 2026

- Built an end-to-end Python simulation of a passive, wirelessly-interrogated capacitive MEMS pressure sensor — modeling the full signal chain from membrane deflection to variable-gap capacitance, LC resonance, and inductive readout.
- Trained a neural-network surrogate on 171 Latin-Hypercube-sampled geometries and applied inverse design to optimize sensor geometry for target performance (~600 times faster than the full simulator, every candidate verified against it) — the same surrogate-screen-then-verify strategy being applied at materials scale in my current ML/DFT based research work.

**Project: Drone** (Funded by FabLab DU) | *Python, Arduino, ESP32* 2025

**Supervised by:** *Dr. Md. Shafiul Alam*

- Co-developed FabLab DU’s first departmental drone for ~USD 66 — integrated propulsion, IMU, and RC link on an ESP32 flight controller, hand-tuned cascaded PID control across all three axes, and resolved ESC desync and vibration instability to achieve 2.8:1 thrust-to-weight and had achieved ~9 min flight time on its first field test.

**ESP32 Retro Gaming Console** | *ESP32* 2026

- Built a handheld console prototype (ESP32, OLED, analog joystick) running three original arcade games with a scrollable menu and persistent high scores — real-time firmware built on millis()-based finite-state machines, ADC filtering with hysteresis, and non-volatile storage across deep-sleep cycles.

**Pocket Reader** — Handheld E-Book Reader | *ESP32* 2026

- Built a self-contained e-book reader prototype serving a mobile-friendly web portal directly from the microcontroller for uploads and Wi-Fi management — no cloud, no external server; on-device REST server with mDNS, LittleFS storage, and deep-sleep power cycling.

**Laser Harp** | *Arduino* 2023

- Academic project from sophomore year — played a two-octave span of musical notes using a laser-and-LED array.

## COMPETENCIES

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**Programming:** Python (scikit-learn, pandas, NumPy, SciPy, SHAP, Matplotlib, Keras/TensorFlow), C/C++, x86 Assembly, Verilog (RTL design), HTML, L<sup>A</sup>T<sub>E</sub>X

**Software & Tools:** VASP, Quantum Espresso, MATLAB, PSPICE, Proteus, AutoCAD, EasyEDA

**Languages:** English (IELTS 7.0; test taken on 2023, outdated now), Bangla (native)

**Soft Skills:** Public Speaking, Team Leadership, Mentoring, Event Management

## REFERENCES

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**Dr. Imtiaz Ahmed**

*Professor, Dept. of EEE*

*Faculty of Engineering and Technology*

University of Dhaka, Dhaka, Bangladesh

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**Dr. Sharnali Islam**

*Associate Professor, Dept. of EEE*

*Faculty of Engineering and Technology*

University of Dhaka, Dhaka, Bangladesh

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*Additional references available upon request.*